

CHE 565 -- Homework 5

Soki Sem

April 24, 2026

Problem 1

Closed-loop Response to Step Disturbance

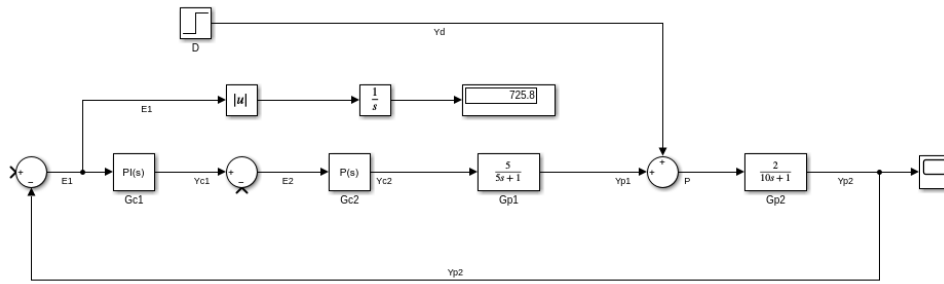


Figure 1: Block diagram of the closed-loop system without cascade control and with a unit step disturbance from Simulink.

The Simulink block diagram is shown in Figure 1.

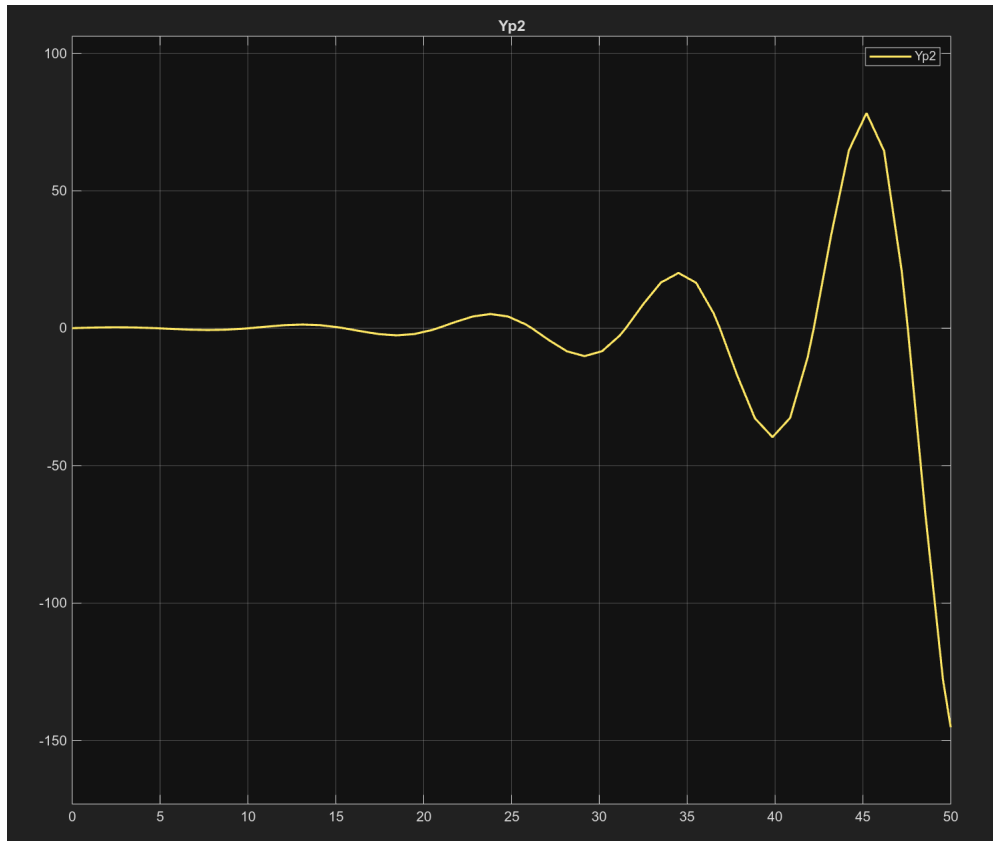


Figure 2: Closed-loop response to no step change in setpoint and a unit step change in disturbance from Simulink.

The Simulink closed-loop disturbance response is shown in Figure 2.

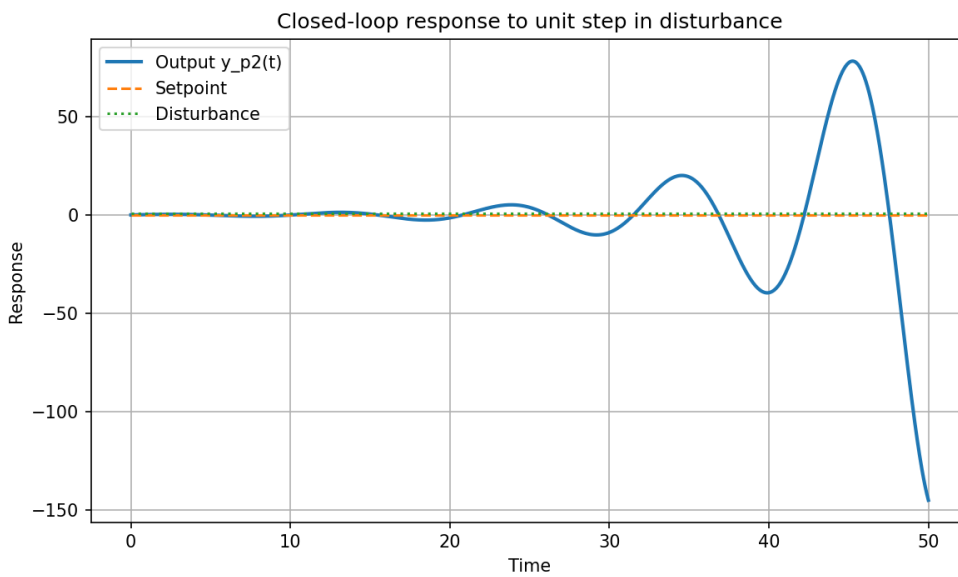


Figure 3: Closed-loop response to no step change in setpoint and a unit step change in disturbance from Python.

The corresponding Python response is shown in Figure 3.

Case 1: Step disturbance with no setpoint change gave $IAE = 725.3696$ in Python and $IAE = 725.8109$ in Simulink.

The PI controller was then tuned using the autotuning feature in Simulink, which gave $P = 0.1837$ and $I = 0.0816$.

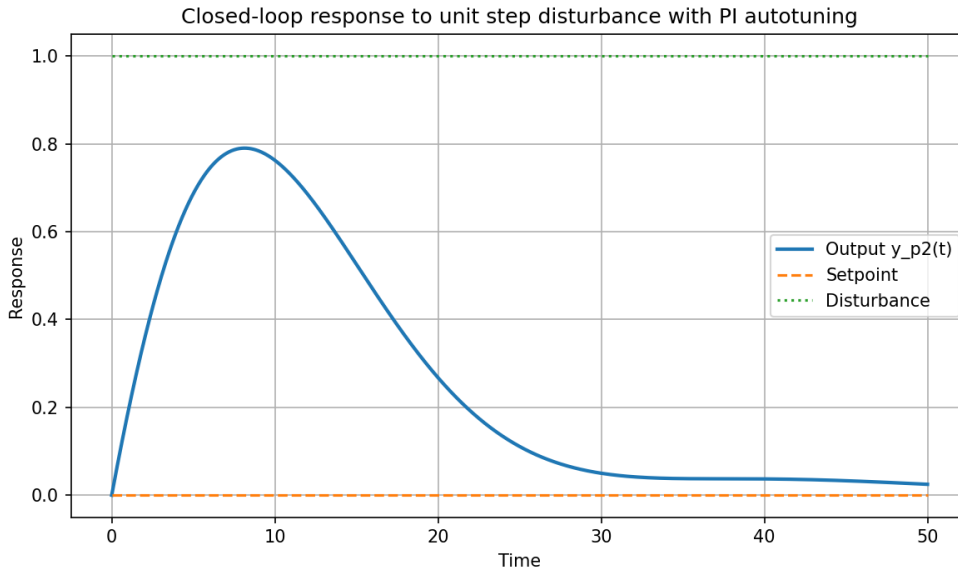


Figure 4: Closed-loop response to no step change in setpoint and a unit step disturbance using the Simulink autotuned PI values in Python.

The Python response using the autotuned PI values is shown in Figure 4.

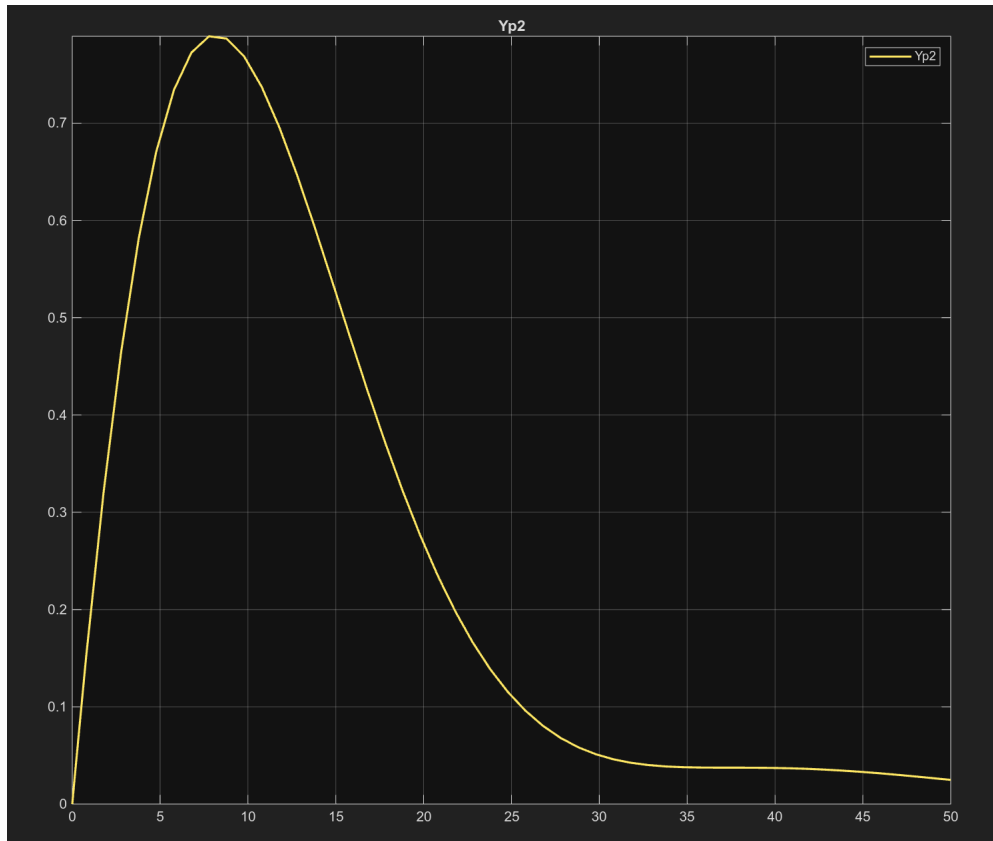


Figure 5: Closed-loop response to no step change in setpoint and a unit step disturbance using PI autotuning from Simulink.

The corresponding Simulink response using PI autotuning is shown in Figure 5.

Case 2: Step disturbance with no setpoint change and PI autotuning gave $IAE = 13.0462$ in Python and $IAE = 13.0463$ in Simulink.

Closed-loop Response to Step Setpoint Change

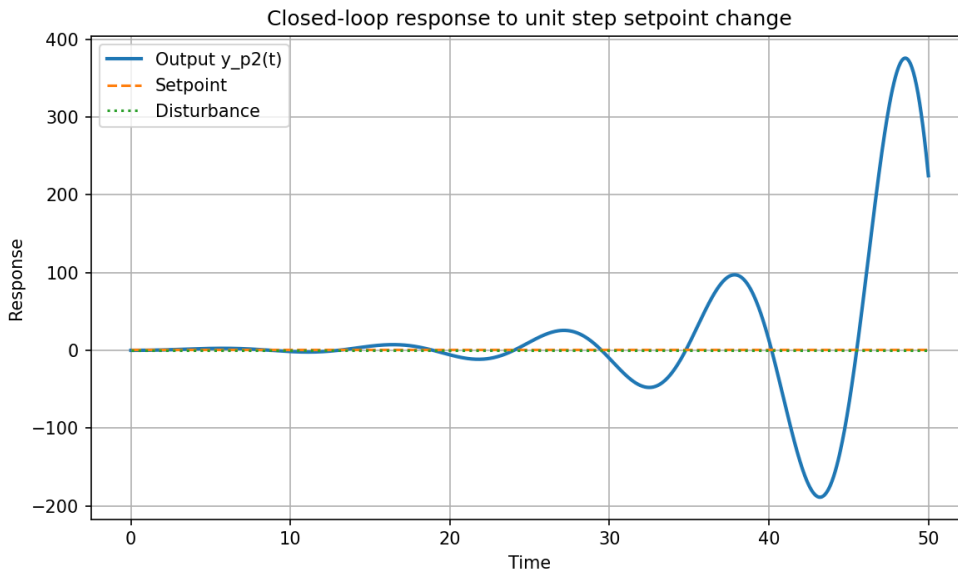


Figure 6: Closed-loop response to unit step setpoint change with no disturbance change from Python.

Next, the response to a unit step change in setpoint with no disturbance change was simulated. The Python response is shown in Figure 6.

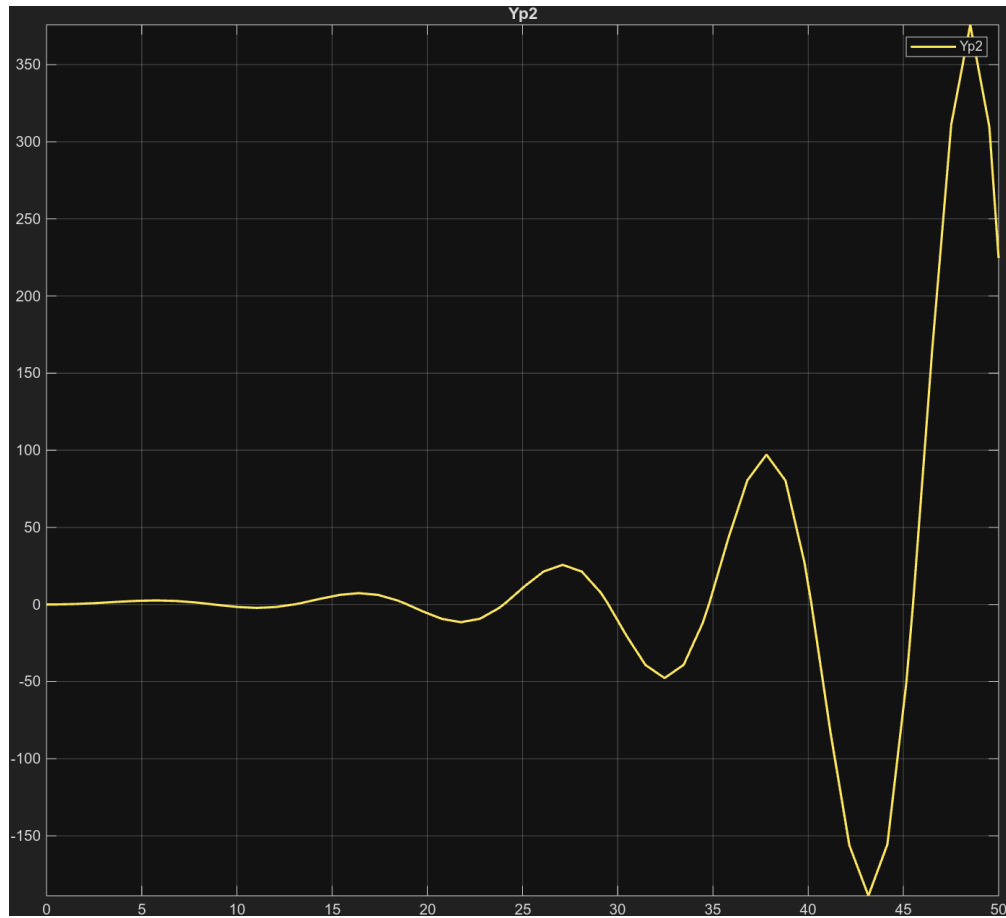


Figure 7: Closed-loop response to unit step setpoint change with no disturbance change from Simulink.

Case 3: Step setpoint change with no disturbance change gave $IAE = 2449.4065$ in Python and $IAE = 2450.8205$ in Simulink.

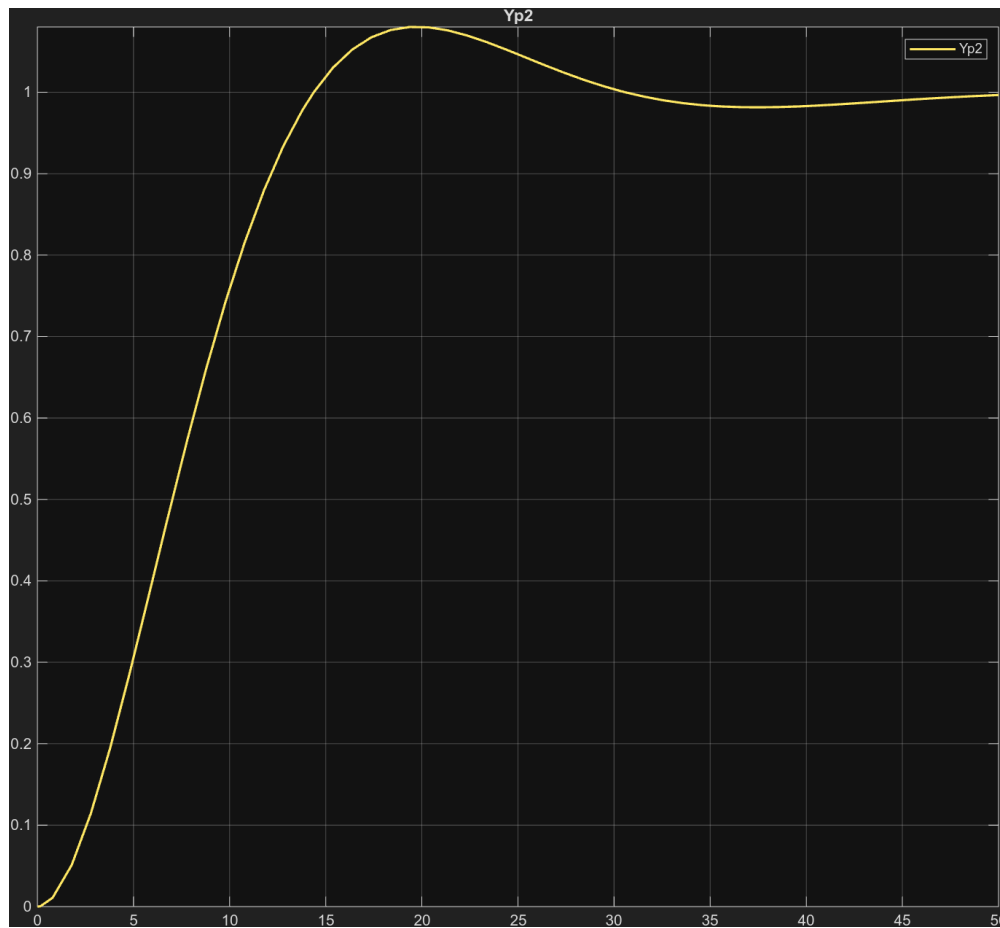


Figure 8: Closed-loop response to unit step setpoint change with no disturbance change from Simulink using PI autotuning.

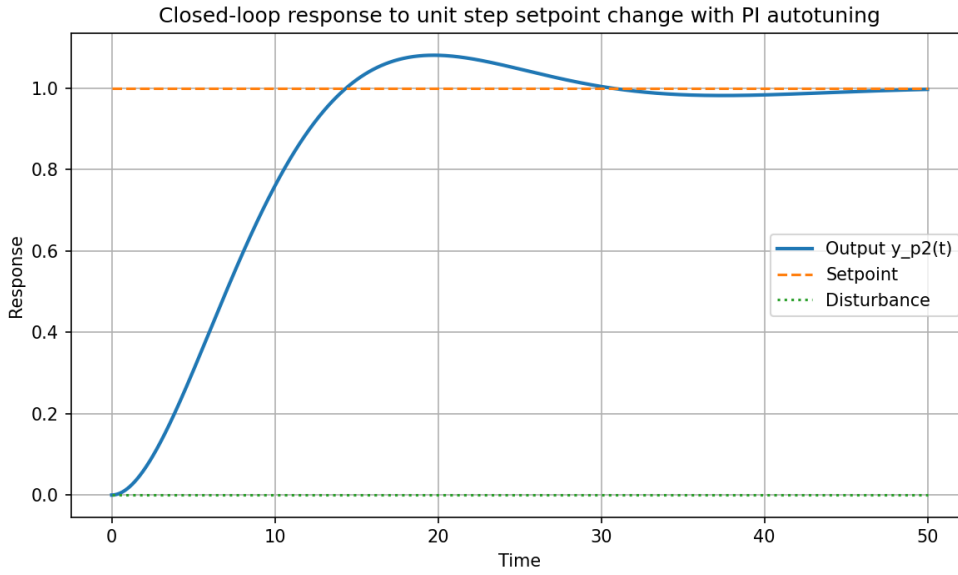


Figure 9: Closed-loop response to unit step setpoint change with no disturbance change from Python using PI autotuning.

The response to a unit step change in setpoint with no disturbance change using PI autotuning is shown in Figure 9.